MATHS ASSIGNMENT

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1. A five digit number is formed by the digits 0, 1, 2, 3, 4 without repetition. Find the probability that the number formed is divisible by 4?

$\Rightarrow$ $S = \{0, 1, 2, 3, 4\}$

The number of 5 digit numbers can be formed using the digits 0, 1, 2, 3, 4 without repetition is $4 \times 4 \times 3 \times 2 \times 1$

$\therefore$ The total possible outcomes, $n = 96$.

Those numbers formed will be divisible by 4 if the two extreme right digits divisible by 4 i.e., the two extreme right digits coul be 4, 12, 20, 24, 32, 40. Now the 5 digit numbers ending in 04, 20 and 40 $= 3 \times 2 \times 1 \times 3 = 18$.

Now the 5 digit numbers ending in 12, 24, 32 $= 2 \times 2 \times 1 \times 3 = 12$.

$\therefore$ The total no. of 5 digit numbers divisible by 4 are $18 + 12 = 30 \Rightarrow m$.

Thus the required probability $= \frac{m}{n}$

$= \frac{30}{96} = \underline{\underline{0.3125}}$

2. A ten digit number is formed using the digits from 0 to 9, every digit being used only once. Find the probability that the number is divisible by 4?

$\Rightarrow$ $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

The no. of 10 digit numbers can be formed using the digits 0 to 9 without repetition is, $n = 9 \times 9!$

Those numbers formed will be divisible by 4, if the two extreme right digits divisible by 4
i.e., the two extreme right digits could be 04, 08, 12, 16, 20, 24, 28, 32, 36, 40, 48, 52, 56, 60, 64, 68, 72, 76, 80, 84, 92, 96.

Now the numbers ending in 04, 08, 20, 40, 60, 80 are equal to $8! \times 6$

Now the numbers ending in 12, 16, 24, 28, 32, 36, 48, 52, 56, 64, 68, 72, 76, 84, 92, 96 = $16 \times 7 \times 7!$

$\therefore$ The total no. of 10-digit numbers divisible by 4 are $(6 \times 8!) + (16 \times 7 \times 7!) = 8,06,400. \Rightarrow m.$

$n = 32,65,920$

Hence the required probability $= \frac{m}{n}$

$= \frac{806400}{3265920}$

$= \underline{\underline{0.2469}}$

3. A bag contains 40 tickets numbered 1, 2, 3, ... 40 of which four are drawn at random and arranged in ascending order $(t_1 < t_2 < t_3 < t_4)$. Find the probability of t_3 being 25.

$\Rightarrow$ 4 tickets can be drawn at random in ${}^{40}C_4$ ways
$\therefore$ The total no. of outcomes, $n = {}^{40}C_4$.

Suppose t_3 is 25, then there are 24 tickets preceeding $t_3 = 25$. The tickets t_1 and t_2 can be choosen from this 24 tickets in ${}^{24}C_2$ ways. Then the ticket t_4 should be greater than 25. Then the no. of ways choosing ticket t_4 from 15 cards $= {}^{15}C_1$.

The no. of favourable outcomes, $m = {}^{24}C_2 \times {}^{15}C_1$

Thus the required probability $= \frac{m}{n}$

$$= \frac{{}^{24}C_2 \times {}^{15}C_1}{{}^{40}C_4}$$

$$= \underline{\underline{0.0453}}$$

4. There are two groups of objects one of which consists of 5 science and 3 engineering subjects, and the other consists of 3 science and 5 engineering subjects. An unbiased die is tossed. If the number 3 or 5 turns up, a subject is selected at random from the first group, otherwise the subject is selected at random from the second group. Find the probability that an engineering subject is selected ultimately?

$\Rightarrow$ Probability of turning up 3 or 5 $= \frac{2}{6} = \frac{1}{3}$.

Probability of selecting and engineering subject from first group $= \frac{{}^3C_1}{{}^8C_1} = \frac{3}{8}$

$\therefore$ Prob. of selecting an engg subject from first group on turning up 3 or 5 $= \frac{1}{3} \times \frac{3}{8} = \frac{1}{8}$

Now, probability of not turning 3 or 5 = $1 - \frac{1}{3} = \frac{2}{3}$.

Probability of selecting an engineering subject from II group = $\frac{{}^5C_1}{{}^8C_1} = \frac{5}{8}$

∴ Prob. of selecting an engg. subject from II group not turning up 3 or 5 = $\frac{2}{3} \times \frac{5}{8} = \frac{5}{12}$.

Thus the probability of selecting an engineering subject is $\frac{1}{8} + \frac{5}{12} = \frac{13}{24}$

5. The odds that a book will be reviewed favourably by three independent critics are 5 to 2, 4 to 3, and 3 to 4. What is the probability that of the three reviews, majority will be favourable?

⇒ The probability that the book shall be reviewed favourably by 1st critic is $\frac{5}{7}$, 2nd critic is $\frac{4}{7}$ and by 3rd critic is $\frac{3}{7}$.

Majority of the 3 reviews will be favourable when two or three are favourable.

Probability that the 1st two are favourable and 3rd is unfavourable = $\frac{5}{7} \times \frac{4}{7} \times \left(1 - \frac{3}{7}\right) = 0.233$

Probability that the 1st and 3rd are favourable and 2nd is unfavourable = $\frac{5}{7} \times \left(1 - \frac{4}{7}\right) \times \frac{3}{7} = 0.131$

Probability that the 2nd and 3rd are favourable and 1st is unfavourable $= \left(1 - \frac{5}{7}\right) \times \frac{4}{7} \times \frac{3}{7} = 0.069$

Finally the probability that all the 3 reviews are favourable $= \frac{5}{7} \times \frac{4}{7} \times \frac{3}{7} = \frac{60}{343} = 0.174$

Since these are mutually exclusive events, then the required probability $= 0.233 + 0.131 + 0.069 + 0.174 = \underline{\underline{0.607}}$

6. In a race, the odds in favour of the four horses H1, H2, H3, H4 are 1:4, 1:5, 1:6, 1:7 respectively. Assuming that a dead heat is not possible, find the probability that one of them wins the race.

$\Rightarrow$ Since it is not possible for all the horses to cover the same distance in the same time and hence the events are mutually exclusive

If P_1, P_2, P_3, P_4 be the probabilities of winning of the horses H_1, H_2, H_3, H_4, then $P_1 = \frac{1}{1+4} = \frac{1}{5}$

similarly $P_2 = \frac{1}{6}$, $P_3 = \frac{1}{7}$ and $P_4 = \frac{1}{8}$.

$\therefore$ the probability that one of them wins the race $= (P_1 \cup P_2 \cup P_3 \cup P_4)$

$= P_1 + P_2 + P_3 + P_4$

$= \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} = \underline{\underline{0.6345}}$

7. A and B throw alternatively with a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins, find his chance of winning.

$\Rightarrow$ The sum 6 can be obtained as follows.
(1,5)(5,1)(2,4)(4,2)(3,3) i.e., 5 ways.
The probability of A is throwing 6 with two dice is, $p_1 = \frac{5}{36}$.

The probability of A is not throwing 6 with two dice, $q_1 = 1 - p_1$

$= 1 - \frac{5}{36} = \frac{31}{36}$

The sum 7 can be obtained as follows.
(1,6)(6,1)(2,5)(5,2)(3,4)(4,3) .i.e., in 6 ways.
The probability of B is throwing 7 with two dice, $p_2 = \frac{6}{36} = \frac{1}{6}$.

The probability of B is not throwing 7 with two dice, $q_2 = 1 - p_2$

$= 1 - \frac{1}{6} = \frac{5}{6}$

Now A can win if he throws 6 in the first, third or fifth .... etc throws.
∴ The chances of A's winning is

$p_1 + q_1 q_2 p_1 + q_1 q_2 q_1 q_2 p_1 + \ldots\ldots$

$p_1 [1 + q_1 q_2 + (q_1 q_2)^2 + \ldots\ldots]$

$$P_1 \left[\frac{1}{1 - q_1 q_2} \right]$$

$$= \frac{5}{36} \left[\frac{1}{1 - \left(\frac{31}{36} \cdot \frac{5}{6} \right)} \right] = \frac{30}{61}$$

8. A and B throw alternatively with a pair of dice. The one who throws 9 first wins. Show that the probabilities of their winning are 9:8.

=> To get 9, the favourable cases are (3,6)(6,3) (5,4)(4,5) i.e., 4 ways.

The probability of getting 9 with two dice by person A, $P_1 = \frac{4}{36}$.

The probability of not getting 9, $q_1 = 1 - P_1$

$$= 1 - \frac{4}{36} = \frac{32}{36}.$$

Let P_2 and q_2 be the probability of getting 9 and not getting 9 with two by person B

$$P_2 = \frac{4}{36}, \quad q_2 = \frac{32}{36}.$$

When A is to win, she should get 9 in first or third or fifth .... throws and B is to win, he should get 9 in second, fourth, sixth.... throws.

∴ the probability of A to win is

$$P_1 + q_1 q_2 P_1 + q_1 q_2 q_1 q_2 P_1 + \ldots\ldots$$

$$P_1 [1 + q_1 q_2 + (q_1 q_2)^2 + \ldots\ldots]$$

$$P_1 \left[\frac{1}{1 - q_1 q_2} \right]$$

$$\frac{4}{36} \left[\frac{1}{1 - \left(\frac{32}{36}\right)^2} \right] = \frac{9}{17}$$

The probability of B to win is

$$P_2 = 1 - \frac{9}{17} \Rightarrow \frac{8}{17}$$

∴ The probability of A to win : prob of B to win

$$\frac{9}{17} : \frac{8}{17} \Rightarrow \underline{\underline{9:8}}$$

9. Let A and B be two events with $P(A) = 1/2$, $P(B) = 1/3$, $P(A \cap B) = 1/4$. Evaluate $P(A/B)$, $P(A \cup B)$, $P(\bar{A}/\bar{B})$.

$$\Rightarrow P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/3} = \underline{\underline{\frac{3}{4}}}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \underline{\underline{\frac{7}{12}}}$$

$$P(\bar{A}/\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{P(\overline{A \cup B})}{1 - P(B)} = \frac{1 - P(A \cup B)}{1 - P(B)}$$

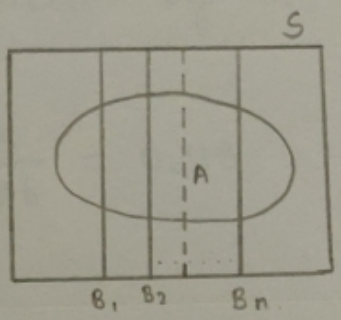
$$= \frac{1 - 7/12}{1 - 1/3} = \underline{\underline{\frac{5}{8}}}$$

10. State and prove Baye's theorem.

=> " If the events $B_1, B_2, \ldots B_n$ be the partitions of sample space 'S', so that their union is S and let A be the any event, then P.T.

$$P(B_i/A) = \frac{P(B_i)\,P(A/B_i)}{\sum_{i=1}^{n} P(B_i)\,P(A/B_i)}$$

Consider the following Venn diagram, From Venn diagram we have.



$$A = S \cap A$$
$$= (B_1 \cup B_2 \cup \ldots) \cap A$$
$$= (B_1 \cap A) \cup (B_2 \cap A) \cup \ldots \cup (B_n \cap A)$$

$$P(A) = P(B_1 \cap A) + P(B_2 \cap A) + \ldots + P(B_n \cap A)$$
$$= P(B_1)\,P(A/B_1) + P(B_2)\,P(A/B_2) + \ldots + P(B_n)\,P(A/B_n)$$

$$P(A) = \sum_{i=1}^{n} P(B_i)\,P(A/B_i)$$

we have $P(B_i/A) = \frac{P(B_i \cap A)}{P(B_i)}$

$$= \frac{P(B_i)\,P(A/B_i)}{\sum_{i=1}^{n} P(B_i)\,P(A/B_i)}$$

11. Three machines M_1, M_2 and M_3 produce identical items. Of their respective output 5%, 4% and 3% of items are defective. On a certain day M_1 has produced 25%, M_2 has produced 30% and M_3 has produced 45%. An item selected at random is found to be defective. What is the probability that it was produced by the machine with the highest output?

=> Let B_1, B_2, B_3 be the events that an item is selected at random is produced by M_1, M_2, M_3 respectively.

$$P(B_1) = \frac{25}{100} = 0.25 \qquad P(B_3) = \frac{45}{100} = 0.45$$

$$P(B_2) = \frac{30}{100} = 0.3$$

Let A be the event of selecting an item taken at random is defective

$$P(A/B_1) = \frac{5}{100} = 0.05 \qquad P(A/B_3) = \frac{3}{100} = 0.03$$

$$P(A/B_2) = \frac{4}{100} = 0.04$$

The highest output from being M_3, the required probability $P(B_3/A) = ?$

$$P(B_3/A) = \frac{P(A/B_3)\,P(B_3)}{P(B_1)\,P(A/B_1) + P(B_2)\,P(A/B_2) + P(B_3)\,P(A/B_3)}$$

$$= \frac{(0.45)(0.03)}{(0.25)(0.05) + (0.3)(0.04) + (0.45)(0.03)}$$

$$= \underline{0.355}$$

12. In a bolt factory there are four machines A, B, C, D manufacturing respectively 20%, 15%, 25% 40% of the total production out of these 5%, 4%, 3%, 2% are defective. If a bolt drawn at random was found defective, what is the probability that it was manufactured by A or D?

=> Let us follow the following notations

Event E: A bolt taken at random is defective.

Event B_1: A bolt taken at random is produced by A

Event B_2: A bolt taken at random is produced by B

Event B_3: A bolt taken at random is produced by C

Event B_4: A bolt taken at random is produced by D.

$P(B_1) = \frac{20}{100} = 0.2$; $P(B_2) = \frac{15}{100} = 0.15$; $P(B_3) = \frac{25}{100} = 0.25$;

$P(B_4) = \frac{40}{100} = 0.4$; $P(E|B_1) = \frac{5}{100} = 0.05$; $P(E|B_2) = 0.04$;

$P(E|B_3) = \frac{3}{100} = 0.03$; $P(E|B_4) = \frac{2}{100} = 0.02$.

$$P(B_1/E) = \frac{P(B_1)P(E|B_1)}{P(B_1)P(E|B_1) + P(B_2)P(E|B_2) + P(B_3)P(E|B_3) + P(B_4)P(E|B_4)}$$

$$= \frac{0.2 \times 0.05}{(0.2 \times 0.05) + (0.15 \times 0.04) + (0.25 \times 0.03) + (0.4 \times 0.02)}$$

$$= \frac{0.01}{0.0315}$$

$$= \underline{\underline{0.3175}}$$

$$P(B_4|E) = \frac{P(B_4)P(E|B_4)}{P(B_1)P(E|B_1) + P(B_2)P(E|B_2) + P(B_3)P(E|B_3) + P(B_4)P(E|B_4)}$$

$$= \frac{0.4 \times 0.02}{0.2(0.05) + 0.15(0.04) + 0.25(0.03) + 0.4(0.02)}$$

$$= \frac{0.4(0.02)}{0.0315}$$

$$= \underline{0.254}$$

Thus, the probability that the defective bolt was manufactured by machine A or machine D

$$= P(B_1/E) + P(B_4|E)$$

$$= 0.3175 + 0.254$$

$$= \underline{0.5715}$$

13. In a certain college, 4% of the boys and 1% of girls are taller than 1.8m. Furthermore 60% of the students are girls. If a student is selected at random and is found to be taller than 1.8m, what is the probability that the student is a girl?

=> Let B_1, B_2 denote the events of selecting a girl student and a boy student respectively.

$P(B_1) = \frac{60}{100} = 0.6$ and $P(B_2) = \frac{40}{100} = 0.4$.

Let A be the event of selecting a student over the height of 1.8m.

$P(A|B_1) = \frac{1}{100}$ and $P(A|B_2) = \frac{4}{100}$

$$P(B_2|A) = \frac{P(B_2)\,P(A|B_2)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2)}$$

$$= \frac{0.6 \times 0.04}{0.01 \times 0.6 + 0.4 \times 0.04}$$

$$= \frac{0.016}{6 \times 10^{-3} + 0.016} = \underline{\underline{0.727}}$$

14. An office has 4 secretaries handling respectively 20%, 60%, 15% and 5% of the files of all the government reports. The probability that the misfile such reports are respectively 0.05, 0.1, 0.1 and 0.05. Find the probability that the misfiled report can be blamed on the first secretary?

=> Let A_1, A_2, A_3 and A_4 be the 4 secretaries of the office respectively handling 20%, 60%, 15% 5% of the files.

$P(A_1) = 0.2$; $P(A_2) = 0.6$; $P(A_3) = 0.15$; $P(A_4) = 0.05$

Let E be the event of misfilling a report by the secretary

$P(E|A_1) = 0.05$; $P(E|A_2) = 0.1$; $P(E|A_3) = 0.1$; $P(E|A_4) = 0.05$

$$P(A_1|E) = \frac{P(A_1)\,P(E|A_1)}{P(A_1)\,P(E|A_1) + P(A_2)\,P(E|A_2) + P(A_3)\,P(E|A_3) + P(A_4)\,P(E|A_4)}$$

$$= \frac{0.2(0.05)}{0.2(0.05) + 0.6(0.1) + 0.15(0.1) + 0.05(0.05)}$$

$\therefore$ $P(A_1/E) = \underline{\underline{0.1143}}$

15. Three machines A, B and C produces 50%, 30% and 20% of the items in a factory. The percentage of defective outputs of these machines is 3, 4 and 5 respectively. If an item is selected at random, what is the probability that it is defective? If a selected item is defective, what is the probability that it is from machine A?

$\Rightarrow$ Let E_1, E_2, E_3 be the events for machines A, B, C

$P(E_1) = \frac{50}{100} = \frac{1}{2}$; $P(E_2) = \frac{30}{100} = \frac{3}{10}$; $P(E_3) = \frac{20}{100} = \frac{1}{5}$

$P(E_1) + P(E_2) + P(E_3) = \frac{1}{2} + \frac{3}{10} + \frac{1}{5} = \frac{5+3+2}{10} = \frac{10}{10} = 1$

events are exhaustive and mutually exclusive.

Let D be the event of item being defective.

$P(D/E_1) = \frac{3}{100}$; $P(D/E_2) = \frac{4}{100}$; $P(D/E_3) = \frac{5}{100}$

$$P(E_1/D) = \frac{P(E_1)P(D/E_1)}{P(E_1)P(D/E_1) + P(E_2)P(D/E_2) + P(E_3)P(D/E_3)}$$

$$= \frac{1/2\,(3/100)}{1/2\,(3/100) + 3/10\,(4/100) + 1/5\,(5/100)}$$

$$= \underline{\underline{\frac{15}{37}}}.$$